Behavioral Responses to Pre-Planned Road Capacity Reduction Based on Smartphone GPS Trajectory Data - A Functional Data Analysis Approach

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ABSTRACT

Pre-planned events such as constructions or special events lead to road capacity reductions and create bottlenecks in the traffic network. The traffic impact of such events goes beyond local areas, as informed drivers may detour to alternative corridors and consequently the traffic congestion may divert or propagate to other corridors. Due to the lack of real observation data, traditional traffic impact analyses are typically based on simulation models, fixed-location sensor data or survey questionnaires. In this research, we use high-resolution vehicle trajectory data collected via a smartphone app, which is capable of keeping track of individual driver's behavior before and after road capacity reduction, to investigate travelers' behavioral responses to pre-planned events and the contribution factors. For this purpose, a functional data analysis (FDA) approach based clustering method is firstly proposed to cluster trajectory data and identify detour patterns, and two logistic and a least absolute shrinkage and selection operator (LASSO) regression models are used to explain drivers' detour behavior choice for each pattern with spatial and temporal features of interest. A case study based on a lane closure event on MoPac expressway in Austin, TX is used as an example in this research. The case study demonstrates that: 1) the freeway capacity reduction triggered heterologous behavior responses, 2) driver detour behavior exhibits three major patterns, and 3) each detour pattern highly depends on spatial features such as trip length, distance to freeway entrance and distance to other alternative freeways, in addition to the temporal features when the trip happens.

Keywords: behavioral change; functional data analysis; preplanned capacity reduction; traffic impact analysis; vehicle trajectory data; smartphone data collection; regression analysis;

1 INTRODUCTION

Pre-planned events such as construction or special events can lead to road capacity reductions and create bottlenecks in the traffic network. The traffic impact of such events goes beyond local areas, as informed drivers may detour to alternative corridors in search for faster travel speeds and consequently the traffic congestion may divert or propagate to other corridors. Due to the lack of real observation data, traditional traffic impact analyses are typically based on simulation models, fixed-location sensor data or survey questionnaires, and is usually assessed at the traffic level, meaning using the aggregated data to measure the change of traffic flows before and after the event. Few prior studies were able to investigate the behavioral responses from individual travelers to such events. The data needed for the driver-level analysis is mostly non-existent.

The wide use of GPS-enabled smartphones and other GPS devices give rise to the new opportunities that allow us to access vehicle trajectory data containing valuable information. Such information includes both travel behavior of a specific person (e.g., origin-destination, departure time, route), and corresponding travel experience (e.g., driving speed, actual travel time, delays). In this research, we use the high-resolution vehicle trajectory data collected via a smartphone app "Metropia", which is capable of keeping track of individual driver's behavior before and after road capacity reduction, to investigate travelers' behavioral responses to pre-planned events and the contribution factors. The objective of this research is to expand the understanding of the impacts of a preplanned road capacity reduction event by studying the heterogeneous behavioral responses at individual driver level.

For this purpose, a functional data analysis (FDA) approach based clustering method is firstly adopted to identify detour patterns by clustering trajectory data. Departs from the commonly seen multivariate data analysis approaches, the proposed FDA approach considers discrete trajectory points as a function of time, thus all the trajectories of different drivers, from different navigation devices, obtained by different sampling frequency, are all modeled uniformly by functions. In other words, under functional data analysis, each trajectory is treated as one functional data. By applying basis expansion techniques such as B-spline expansion, Fourier transformation and wavelet decomposition, each raw trajectory data can be modeled and expressed in a functional form. With such data analysis approach, all trajectories which sampled in different scales (e.g. some data GPS sampled every 1 second but others sampled every 2 second) can be uniformly expressed in the same functional form. Additional benefits of such approach also include the reduction of unnecessary noise in trajectory data by basis expansion smoothing.

With the detour pattern identified, drivers' detour behavior choice for each pattern is explained by two logistic and a least absolute shrinkage and selection operator (LASSO) multinomial regression models. Temporal features of interest, including system-estimated travel time, actual travel time, departure time, day of the week, and spatial features including trip length, distance between origin/destination and capacity reduction site, distance between original/destination and alternative corridors, and minimum distace from capacity reduction site to origin or destination, are defined and used in this modeling process. A case study based on a lane closure event on MoPac expressway in Austin, TX is used as an example in this research.

The remainder of this paper is organized as follows: Section 2 reviews the relevant literature in the past and introduces background knowledge about function data analysis approach. Section 3 briefly talks about the data preparation. Section 4 documents the modeling process and the

spatial and temporal features defined, and section 5 presents the case study process and results. Section 6 reviews and concludes this research.

2 LITERATURE REVIEW

2.1 Traffic impact analysis

The study of roadway capacity reduction impact draws the considerable attention of researchers and practitioners. A popular area of research interest is the accurate estimation of freeway capacity as the result of vehicle crashes and driving behavior at the incident location, work zone activities, severe weather, and so on (Hall and Agyemang-Duah 1991, Ober-Sundermeier and Zackor 2001, Ahmed and Fred 2003, Brian et al. 2003, Smith et al. 2004, Agarwal et al. 2006, Victor et al. 2008). In the event that capacity reduction degree is known or could be assumed, simulation models are frequently used to analyze the traffic impacts of road capacity reduction at the network level. For example, Mahmassani and Jayakrishnan adopted a dynamic simulation model that has a macroparticle traffic simulator, and a user decision component to represent traffic system disruptions in the form of lane closures (Hani and Jayakrishnan 1988). Mitra proposed a framework to evaluate the impact of variable speed limits on work zone traffic operation with Vissim (Mitra and Pant 2005), while Li et al. used PARAMICS to evaluate the control strategies of variable speed limits (Li et al. 2011). More studies applying a simulation approach can be found (Nemeth and Rathi 1985, Washburn et al. 2008, Watling et al. 2012). These studies, although capable of revealing system-wide traffic changes, are not based on real data and certain assumptions have to be made for the simulation model to approximate real-world traffic conditions. For example, it's generally known that the calibration of a model, particularly O-D demand calibration, plays a significant role in the scenario analysis, and depending on the system setup, travelers may respond to the capacity reduction events differently.

Another network level analysis approach that is frequently seen in the literature is based on the traditional fixed-location sensor data. For example, Zhu et al. used loop detector data to analyze the traffic and behavioral effects of the I-35W Mississippi River bridge collapse (Zhu et al. 2010). Also by using the dataset collected by loop detectors, He and Liu modeled the day-to-day traffic evolution process after unexpected network disruption (He and Liu 2012), Xie and Levinson modeled the economic loss incurred by increased travel delay in alternative scenarios (Xie and Levinson 2011), and Lee et al. analyzed the impacts of urban freeway rehabilitation on network traffic (Lee et al. 2004). In addition, Victor et al. utilized high-quality videos of the traffic flow around two accidents recorded from a helicopter to calibrate the capacity reduction(Victor et al. 2008). Such analysis based upon fixed-location sensor data is able to measure the impact of the capacity reduction in a quantifiable way, however, due to the nature of the fixed-location sensor, the individual traveler's behavior change cannot be captured with this approach, nor can the reasons for the detour behavior be revealed.

Besides the system-level analysis, understanding driver behavior, including how travel behavior changes due to road capacity reduction, and what the underlying features that determine behavior change at the individual level are will help agencies better cope with road incidents when they occur. To achieve this goal, the survey-based approach is frequently used. For example, Ye et al. (Ye et al. 2012) analyzed the impact on commuter's behavior during the 'Fix I-5' project in California based on two waves of survey data. Nam et al. (Nam et al. 1999) analyzed the travel pattern changes during pavement of I-405 in Seattle. Fujii et al. (Fujii et al. 2001) surveyed 335 drivers in Japan, which revealed that temporary freeway closure has a significant impact on the driver's perception of commute time thereby influencing travel behavior decisions. Although it is

a step further towards understanding traveler's behavior changes, the survey-based approach is not based on observation and it also suffers from various problems such as miss reporting and nonresponse bias.

2.2 Function data analysis approach overview

Investigation on travel behaviors or patterns from trajectory data can be found in GIS studies. For large amount of trajectory data, it is difficult to visually learn behavior patterns because they are overlapped with each other on the map. To address this problem, trajectories clustering becomes a promising solution as summarized by Morris and Trivedi (2009). Most of these methods extract different features from the trajectories for clustering, such as applying principal component analysis (PCA) to automatically extract features from the trajectories data (Bashir et al. 2007). However, this approach works well on images but not for GPS travel trajectories as trajectories' duration and distance might differ from each other significantly. Some long trip trajectories might have thousands of GPS points while others might only have a few hundred. It is impossible to apply PCA to trajectory points directly because of the dimension inconsistent.

To process this kind of "curve-liked" data, one advanced and popular method is functional data analysis (FDA) (Ramsay and Silverman 2005). Departs from the commonly seen multivariate data analysis approaches, the proposed FDA approach considers discrete trajectory points as a function of time, thus all the trajectories of different drivers, from different navigation devices, obtained by different sampling frequency, are all modeled uniformly by functions. In other words, under functional data analysis approach, each trajectory is treated as one functional data. By applying basis expansion techniques such as B-spline expansion denoted in Eq.(1), each raw trajectory data can be modeled and expressed in a functional form.

$$y(t) = \sum_{j=1}^{M} \beta_j \varphi_j(t) \tag{1}$$

where:

- $y(t)$ is the original function,
- $\varphi_j(t)$ is j^{th} basis functions,
- β_j is the coefficient of the corresponding basis function.

With such data analysis approach, all trajectories which sampled in different scales (e.g. some data GPS sampled every 1 second but others sampled every 2 seconds) can be uniformly expressed in the same functional form. Additional benefits of such approach also include the reduction of unnecessary noise in trajectory data by basis expansion smoothing.

Based on this model, all the information from the raw data can be projected to M basis coefficients β_j. Obtaining the basis coefficients can be done through a least square regression, this process is also known as B-spline smoothing. Once trajectory data are presented as functions, cluster methods can be applied. Section 4.1 below will further describe the basis expansion and modeling process.

3 DATA PREPARATION

3.1 Data description

Detailed trajectory data and driving behavior data for each trip are collected through Metropia smartphone app. When a user starts a trip with the app, the internal GPS module built in the smartphone is activated and starts to record the second-by-second data. These data, including detailed position such as latitude, longitude velocity and acceleration will be collected at fine time interval and sent back to the cloud server, where they will be stored and used for further analysis.

Figure 1 gives a visualization of the collected trajectory data. Following is a detailed list of attributes of the trajectory data.

- UserID: The unique ID of the user
- TripID: The unique ID of the particular trip
- Latitude: The latitude of this particular GPS point
- Longitude: The longitude of this GPS point
- Speed: The instantaneous travel speed of the vehicle
- Timestamp: The timestamp of this GPS point, with1 second data collection interval
- LinkID: The ID of the road segment that user is currently driving on
- System suggested route: a list of links suggested by the system
- System predicted travel time: the estimated travel time predicted by the system

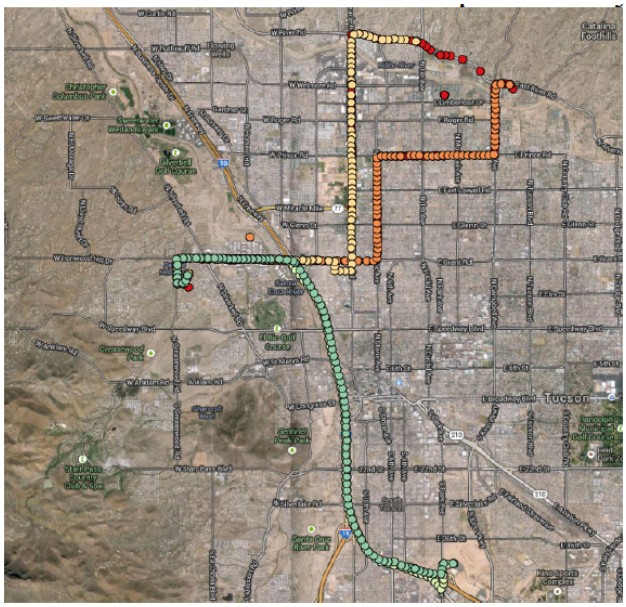


Figure 1 Smartphone trajectory data visualization

3.2 Case study setup

A lane closure event on MoPac expressway, Austin, Texas, which started from Feb. 21 2016, and temporarily reduced MoPac northbound from three lanes to two lanes between Lady Bird Lake and Enfield Road, is used as a case study to analyze traveler's behavioral responses to capacity reduction events. MoPac expressway, also called Texas state highway loop 1, is one of the key arteries across the downtown area of Austin. Figure 2 shows an overview of the network and the bottleneck location (black arrow) where the capacity reduction happens.

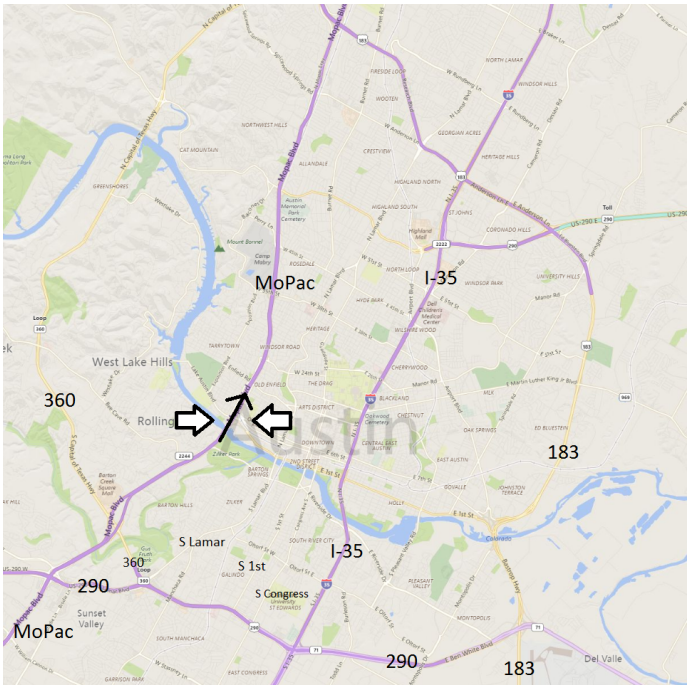


Figure 2 Overview of case study setup

Regarding the trajectory data collected, before the lane closure event, a total of 1,543 trips from 261 users were identified to have travelled through the capacity reduction site, while only 306 such trips from 98 users were found after the event. Additionally, 15,365 trips from 514 users were collected that went along the same direction but did not travel on MoPac either before or after the lane closure event, in other words, they always preferred alternative routes. We did not impose constraints on user's attributes in this data selection process, as while methodologically it is possible to infer them such as trip purpose from the trajectory data (Zhu et al. 2018), the data training process requires each user have significant amount of trajectory data, and would reduce sample size significantly if those criteria are considered.

A few criteria were used to identify the trip pairs that went through the capacity reduction site both before and after: 1) same user ID, 2) same day of the week, 3) departure time within an hour, 4) original locations within a quarter mile, and 5) destination locations within a quarter mile. After applying such matching criteria, a total of 539 trip pairs were identified to have continued to travel through MoPac construction location.

4 STATISTICAL MODELING METHODOLOGY

4.1 Basis expansion and functional data structure

Let's first define the following notations:

- i denotes the trip id, i=1, 2, …, N
- k denotes k^{th} observed GPS points, k=1, 2, …, K_i
- g_{ik} as the kth GPS point data collected in trip i
- $t_{i,k}$ is scaled time point of kth observation in trip i
- $lat_{g_{ik}}$ and $lon_{g_{ik}}$ as the latitude and longitude of gps point g_{ik}, respectively.

Since trajectory data obtained by navigation system is a set of longitude and latitude points, it can be represented by $(\mathrm{t}_{i,k}, lon_{i,k}, lat_{i,k})$. Because for different trip i, the number of observed GPS points are different, K_i is used to denote the number of trajectory points for trip i.

Based on B-spline expansion, these discrete points can be modeled by a continues function:

$$\mathrm{lon_i}(t) = \sum_{j=1}^{M} \alpha_{i,j}\varphi_j(t) = \begin{bmatrix}\alpha_{i,1} & \alpha_{i,2} & \dots & \alpha_{i,M}\end{bmatrix}\begin{bmatrix}\varphi_1(t)\\ \varphi_2(t)\\ \dots\\ \varphi_M(t)\end{bmatrix} = \boldsymbol{\alpha}_i^T\boldsymbol{\varphi}(t) \tag{2}$$

$$\mathrm{lat_i}(t) = \sum_{j=1}^{M} \beta_{i,j}\varphi_j(t) = \begin{bmatrix}\beta_{i,1} & \beta_{i,2} & \dots & \beta_{i,M}\end{bmatrix}\begin{bmatrix}\varphi_1(t)\\ \varphi_2(t)\\ \dots\\ \varphi_M(t)\end{bmatrix} = \boldsymbol{\beta}_i^T\boldsymbol{\varphi}(t) \tag{3}$$

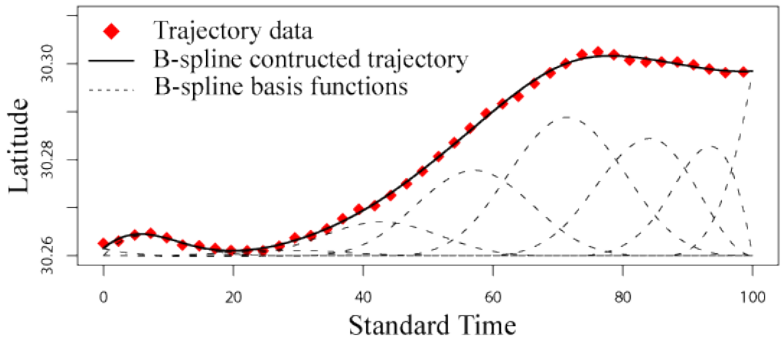


Figure 3 Illustration of B-Spline Expansion

An example of B-spline expansion is depicted in Figure 3, where a smoothed trajectory function (solid black curve) is represented as summation of B-spline basis functions (dashed black curves) to model the raw trajectory data (red diamond). The height of these basis functions are determined by the basis coefficients $\boldsymbol{\alpha}_i^T$ and $\boldsymbol{\beta}_i^T$. Such basis expansion method is advantageous in terms of transferring massive amount of GPS trajectory data points into several basis functions' coefficients $\boldsymbol{\alpha}_i^T$ and $\boldsymbol{\beta}_i^T$ without losing trajectory pattern.

To obtain the basis coefficients, a least square regression model is constructed as following:

$$lonSSE_i = \sum_{k=1}^{K_i}\left[lon_{i,k} - \mathrm{lon_i}(\mathrm{t}_{i,k})\right]^2 = \sum_{k=1}^{K_i}\left[lon_{i,k} - \boldsymbol{\alpha}_i^T\boldsymbol{\varphi}(\mathrm{t}_{i,k})\right]^2 \tag{4}$$

$$latSSE_i = \sum_{k=1}^{K_i}\left[lat_{i,k} - \mathrm{lat_i}(\mathrm{t}_{i,k})\right]^2 = \sum_{k=1}^{K_i}\left[lat_{i,k} - \boldsymbol{\beta}_i^T\boldsymbol{\varphi}(\mathrm{t}_{i,k})\right]^2 \tag{5}$$

To simplify the notations for the lease square model, some matrix-formed data are introduced as following:

$$\boldsymbol{lon}_i = \begin{bmatrix}lon_{i,1}\\ lon_{i,2}\\ \dots\\ lon_{i,K_i}\end{bmatrix};\ \boldsymbol{lat}_i = \begin{bmatrix}lat_{i,1}\\ lat_{i,2}\\ \dots\\ lat_{i,K_i}\end{bmatrix};\boldsymbol{\Phi}_i = \begin{bmatrix}\varphi_1(t_{i,1}) & \varphi_2(t_{i,1}) & \cdots & \varphi_M(t_{i,1})\\ \varphi_1(t_{i,2}) & \varphi_2(t_{i,2}) & \dots & \varphi_M(t_{i,2})\\ \vdots & \vdots & \ddots & \vdots\\ \varphi_1(t_{i,K_i}) & \varphi_2(t_{i,K_i}) & \cdots & \varphi_M(t_{i,K_i})\end{bmatrix} \tag{6}$$

where $\boldsymbol{lon}_i$ and $\boldsymbol{lat}_i$ are vectors which contain all longitude and latitude points. $\boldsymbol{\Phi}_i$ is a K_i×M matrix; each column is the basis function value at all the time points. By reconstructing these trajectory data, the least square model can be rewritten in a simple quadratic form:

$$lonSSE_i = (\boldsymbol{lon}_i - \boldsymbol{\Phi}_i\boldsymbol{\alpha}_i)^T(\boldsymbol{lon}_i - \boldsymbol{\Phi}_i\boldsymbol{\alpha}_i) \quad (7)$$

$$latSSE_i = (\boldsymbol{lat}_i - \boldsymbol{\Phi}_i\boldsymbol{\beta}_i)^T(\boldsymbol{lat}_i - \boldsymbol{\Phi}_i\boldsymbol{\beta}_i) \quad (8)$$

Thus the basis coefficients for trajectory $\underline{i}$ can be estimated by:

$$\widehat{\boldsymbol{\alpha}}_i = (\boldsymbol{\Phi}_i{}^T\boldsymbol{\Phi}_i)^{-1}\boldsymbol{\Phi}_i{}^T\boldsymbol{lon}_i \quad (9)$$

$$\widehat{\boldsymbol{\beta}}_i = (\boldsymbol{\Phi}_i{}^T\boldsymbol{\Phi}_i)^{-1}\boldsymbol{\Phi}_i{}^T\boldsymbol{lat}_i \quad (10)$$

Through B-spline model and least square regression, all the trajectories information is converted into the basis coefficients. The longitude latitude functions can be obtained by: $\widehat{\text{lon}}_i(t) = \widehat{\boldsymbol{\alpha}}_i^T\boldsymbol{\varphi}(t)$; $\widehat{\text{lat}}_i(t) = \widehat{\boldsymbol{\beta}}_i^T\boldsymbol{\varphi}(t)$.

The result of basis expansion is illustrated in Figure 4, where the black solid curves are observed trajectory data and the red dashed curves are based on B-Spline expansion. Raw GPS trajectory data consist of the longitude values and latitude values associated with Unix time. As illustrated in Figure 4, there are two curves to represent the trajectories temporally and spatially: longitude curve and latitude curve. All trips start from different time and might have different length. To make these trajectories comparable in curve shape perspective, all the unix time are standardized into the same scale. In this example, there are about 3000 data points in observed raw curve, while for B-spline expansion curves, there are only 20 B-splines. Which means that B-Spline expansion reduced the dimension of trajectory curves greatly without loss much information of the curves. In the other word, Basis expansion captures most features of the curve with only 20 coefficients.

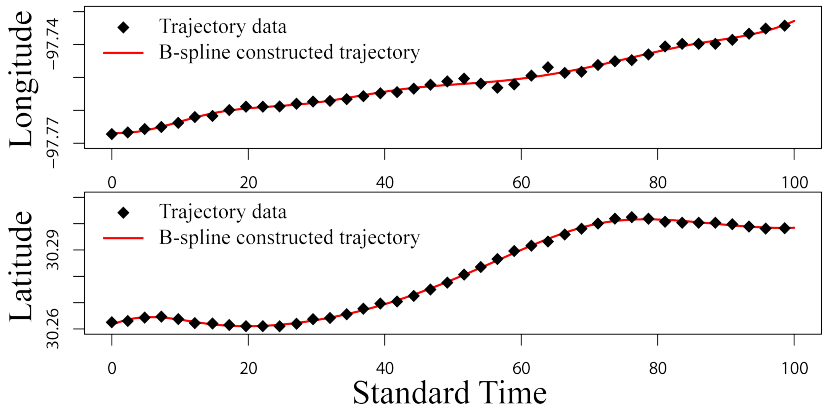


Figure 4 Visualization of trajectory data by B-Spline expansion

4.2 Features definition

To identify and explain the main factors influencing detour decisions, statistical models are built. This section describes the spatial and temporal features defined for such modeling process.

Spatial features definition

A total of four spatial features are defined to characterize a trip from the spatial dimension, including trip length, distance between origin/destination and capacity reduction site, distance

between original/destination and alternative corridors, and minimum distance from capacity reduction site to origin or destination.

These spatial features are defined as the logarithm of distance. By using the logarithm, the linear combinations of spatial features become the proportional relationship between distances. As distance proportion is of interest in this study, the logarithm is necessary here, which makes linear additive models such as logit regression applicable.

If we further denote:

- cr as the capacity reduction site,
- c_{ip} as the pth alternative corridor associated with trip i.
- $dist\,\{g(i,k), g(i,k')\}$ as the distance between gps point g_{ik} and $g_{ik'}$, and Manhattan distance is used to approximate the distance between two gps points.
- G as the number of GPS points in trip i

Let S_{i1} be the logarithm of trips' OD distance for trip i, we will have:

$$S_{i1} = \log \sum_{k=0}^{G-2} dist\,\{g(i,k), g(i,k+1)\} = \log \sum_{k=0}^{G-2} \sqrt{(lat_{g_{ik}} - lat_{g_{i(k+1)}})^2 + (lon_{g_{ik}} - lon_{g_{i(k+1)}})^2} \quad (11)$$

Let S_{i2} be the logarithm of distance from the capacity reduction site to origin and destination for trip i, we will have:

$$\begin{aligned} S_{i2} = \log[&dist\,\{g(i,0), cr\} + dist\,\{g(i,G-1), cr\}] \\ &= \log[\sqrt{\left(lat_{g_{i0}} - lat_{cr}\right)^2 + \left(lon_{g_{i0}} - lon_{cr}\right)^2} \\ &+ \sqrt{\left(lat_{g_{i(G-1)}} - lat_{cr}\right)^2 + \left(lon_{g_{i(G-1)}} - lon_{cr}\right)^2}] \end{aligned} \quad (12)$$

Let S_{i3} be the logarithm of distance from other alternative corridors to origin and destination for trip i, we will have:

$$\begin{aligned} S_{i3} = min \log[&dist\left\{g(i,0), c_{ip}\right\} + dist\left\{g(i,G-1), c_{ip}\right\}] = \\ min \log[&\sqrt{\left(lat_{g_{i0}} - lat_{c_{ipl}}\right)^2 + \left(lon_{g_{i0}} - lon_{c_{ipl}}\right)^2} + \\ &\sqrt{\left(lat_{g_{i(G-1)}} - lat_{c_{ipl}}\right)^2 + \left(lon_{g_{i(G-1)}} - lon_{c_{ipl}}\right)^2}] \\ &\text{for any } p \in P \text{ and } c_{ipl} \in c_{ip} \end{aligned} \quad (13)$$

Let S_{i4} be the logarithm of minimum distance from capacity reduction site to origin or destination for trip i, we will have:

$$S_{i4} = min\,(\log[dist\,\{g(i,0), cr\}, dist\ log\{g(i,G-1), cr\}])$$

$$= \frac{\min\left(\log\sqrt{\left(lat_{g_{i0}} - lat_{cr}\right)^2 + \left(lon_{g_{i0}} - lon_{cr}\right)^2},\right.}{\left. log\sqrt{\left(lat_{g_{i(G-1)}} - lat_{cr}\right)^2 + \left(lon_{g_{i(G-1)}} - lon_{cr}\right)^2}\right)} \quad (14)$$

Temporal features definition

A total of four temporal features are defined to characterize a trip from the temporal dimension, including actual trip travel time, system-estimated travel time, departure time, day of the week.

Let T_{i1} be the actual travel time for trip i, T_{i2} be the system-estimated travel time before user starts to travel, T_{i3} be the day of the week, and T_{i4} be the departure time flag for trip i.

we will have:

$$T_{i1} = t_{g_{i0}} - t_{g_{i(G-1)}} \text{where } t_{g_{ij}} \text{ as the timestamp associated with } g_{ij} \quad (15)$$

$$T_{i3} = \begin{cases} 0 \text{ if } \mathrm{t_{g_{ik}}} \text{ belongs to weekday} \\ 1 \text{ if } \mathrm{t_{g_{ik}}} \text{ belongs to weekday} \end{cases} \quad (16)$$

$$T_{i4} = \begin{cases} 0 \text{ if } \mathrm{t_{g_{ik}}} \text{ belongs to off peak hours} \\ 1 \text{ if } \mathrm{t_{g_{ik}}} \text{ belongs to peak hours } (7-9\text{am}, 16-18\text{pm}) \end{cases} \quad (17)$$

Detour variables

Let Y_{i1} be a binary variable denoting if a user has detoured due to the capacity reduction event for trip i. We will have

$$Y_{i1} = \begin{cases} 0 \text{ if user stays on the same route} \\ 1 \text{ if user detoured from the route} \end{cases} \quad (18)$$

Let Y_{i2} denote a discrete variable that describes the detour pattern choice. Depend on however many patterns can be identified from the clustering analysis, we will have

$$Y_{i2} = \begin{cases} 0 \text{ if user stays on the same route} \\ 1 \text{ if detour pattern 1 is observed} \\ p \text{ if detour pattern } p \text{ is observed} \end{cases} \quad (19)$$

5 BEHAVIORAL CHANGE ANALYSIS

In this section, we first briefly analyze the outcomes associated with the detour behaviors, and identify major detoured patterns based on the methodologies defined earlier. After that, the main reasons affecting driver's detouring behavior is further examined by statistic models.

5.1 Detour outcome analysis

Detour outcome analysis aims to provide an overview of behavioral outcomes comparison among those who detoured versus those stayed, and provides insights on the detour reasons at an aggregated level. Such analysis focus on two aspects, travel time changes and travel time reliability.

Regarding travel time changes, analysis results for 539 trip pairs that continue to use MoPac after lane closures indicate that:

1. On average, travel time increased by 5.7 minutes per trip pair, which translates into a 27% increases overall.
2. For 428 out of 539 trip pairs, or 79% of the trips, actual travel time after lane closure is longer than before.
3. For the trips that avoided the MoPac construction site, travel times averaged 7.0 min longer than before, which translates into a travel time increase of 31%.

These numbers reveal a somehow surprising fact that although road construction is ongoing, MoPac remained to be the best corridor to travel for the northbound traffic in Austin. This is probably due to the location of MoPac which makes it still a better option than others if users were to detour to a longer route.

Travel time prediction accuracy, i.e. how accurate can the travel time be predicted over a certain route, is used to represent the travel time reliability. By comparing the system estimated travel time with the driver's experienced travel time, it is found that on average

1. The predicted travel time underestimates the actual travel time by 12.8% for the stay trips, and
2. The predicted travel time underestimates by 8% for the detour trips.

These results indicate that MoPac traffic conditions during the construction related lane closure were less stable and more difficult to predict when compared with other corridors. The differences in prediction accuracy in a sense may be able to explain the travel behaviors of the detoured drivers. That is, although detouring to alternate routes may take more time than staying on MoPac, the travel time may be more reliable, and therefore makes the detour trips attractive.

It is interesting to observe that drivers sometimes prefer to detour from MoPac to other alternative routes, even though they may experience a longer travel time. In the next sections, we further investigate the patterns of users making detours and the main reasons affecting drivers' detour behavior.

5.2 Detour pattern identification

Based on the FDA approach method presented in section 4.1, trajectory data are clustered via Gaussian mixture model (GMM). After representing trajectory data of each trip via functional data analysis approach with a set of B-Spline basis functions, as discussed in section 4.1, each trip would come with coefficients $\boldsymbol{\alpha}_i^T$ and $\boldsymbol{\beta}_i^T$. GMM is then applied to cluster trips based on their characteristics described by the coefficients $\boldsymbol{\alpha}_i^T$ and $\boldsymbol{\beta}_i^T$. Figure 5 illustrated the three types of detour patterns identified.

The first detour pattern is the group of drivers who have options to use a local arterial, such as Lamar Blvd, to avoid freeway traffic and the MoPac construction site. These drivers usually originate from somewhere south of the river to go to downtown Austin or further north to destinations located between MoPac and I-35. It can be noticed that these trips are typically short. The suspicion is that for short trips, users are more willing to take a short detour just to avoid that particular construction area.

The second detour pattern is the group of drivers who shifted to other alternative freeways or highways. They could use TX-360, I-35, TX-183, or TX-130 to avoid the construction zone on MoPac. It is noticed that this kind of trips are usually pretty long, indicating for the longer trips, drivers may have more options on choosing which freeway to travel, and thus were able to avoid taking MoPac expressway.

The third detour pattern is the group of drivers who either have origin or destination addresses located downtown Austin. For these trips, most drivers choose to avoid the construction zone by leaving MoPac earlier or entering MoPac later, while some other drivers choose to use I-35 or other corridors.



(1): Avoid Freeway (2): Use other freeways (3): O or D in downtown

Figure 5: Visualization of three detour patterns

While detour patterns were able to be identified through the clustering method, the underlying reasons were not clear and needs to be analyzed. The next section will build statistical models to explain what the attributes influencing the detour decision are.

5.3 Detour behavior choice modeling

Regression models are employed to further explore the relationship between these features and detour choice. First of all, 2 logit models are applied to model these features. The first model is a binomial logistic regression model, where the response variable is Y_1 as defined in section 4.2. The second model is a multinomial logistic regression; in this model the response is Y_2 as defined in section 4.2. The results and significant level of these 2 models are demonstrated in Table 1 below.

Table 1 Coefficients and significant level of logit models

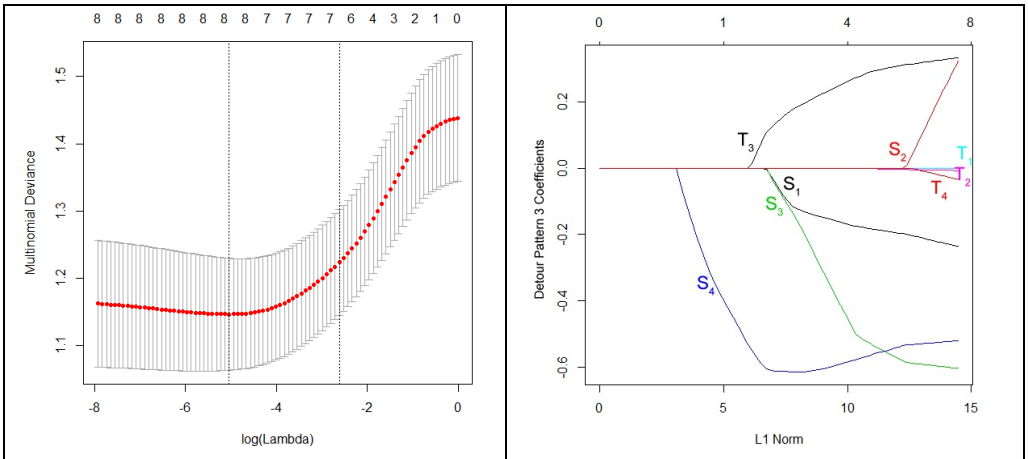
Features	Model 1	Model 2		
	Detour Coefficient	Detour Pattern 1 Coefficient	Detour Pattern 2 Coefficient	Detour Pattern 3 Coefficient
Intercept	0.8202	7.0439 ***	-5.0563 ***	-1.5548 *
S_1	-1.6125 ***	-4.9128 ***	-0.1852	-0.6808
S_2	1.3437 ***	0.0921	1.5143 ***	0.8443 **
S_3	-0.6421 *	-1.8927 **	0.6425	-1.2145 **
S_4	-0.3531 *	-0.4997	0.1088	-0.7622 ***
T_1	0.0731 ***	0.0833 **	0.1046 ***	0.0494 **
T_2	-0.0757 ***	-0.0706	-0.0976 ***	-0.0382

T_3	0.1756	-0.3332	0.0836	0.3796
T_4	-0.6201	-0.9478	-0.3663	-0.4460
Model Evaluation				
AIC	600.97	805.4908		
PseudoR^2	0.2496	0.4027993		
Significance level: * 0.1; ** 0.05; *** 0.01				

These two models tell us that all four spatial features are significant for driver detour decision-making. As for the temporal features, the estimated travel time and actual travel time appear to be significant. Day of the week and time of day are less important when compared with travel time.

However, the significance level of the variables shown in Model 2 is not sufficient for variable selection here. Firstly, some of the features such as travel duration and estimated duration might be highly correlated, which will result in a multicollinearity problem. Secondly, logistic regression includes all features simultaneously, while variable selection is usually a forward or backward procedure, and variable removal or addition will change the whole model every step of the way. The logit models have the limitation to screen out variables which are less influential.

To overcome the above-mentioned issues, select proper variables and build a robust model, LASSO multinomial regression is applied. Based on the previous analysis, a multinomial model is more adequate, and LASSO, is one of the most popular methods for variable selection. A 10-fold crossvalidation of the LASSO model is shown in Figure 6-a, where the upper bound of cross-validation error indicates that 7 is the minimum number of variables. The significance of features can be determined by the LASSO path.



(a) 10-fold Cross-Validation of LASSO (b) LASSO Path for Detour Pattern 3

Figure 6 LASSO regression illustration

Further, Figure 6-b shows an example of the LASSO path for detour pattern 3. The features are ordered by their significance as: $S_4 > T_3 > S_3 = S_1 > S_2 > T_4 > T_2 > T_1$. Based on cross validation and LASSO path, the final model is obtained as shown in Table 2.

Table 2 LASSO model results

Features	LASSO Model		
	Detour Pattern 1 Coefficient	Detour Pattern 2 Coefficient	Detour Pattern 3 Coefficient
Intercept	5.0318	-3.2654	-2.1215
S_1	-4.0005	0.0858	-0.0858
S_2	0	0	0
S_3	-0.0396	0.0396	-0.0397
S_4	0	0	-0.5608
T_1	0	0.0288	0
T_2	0	-0.0320	0
T_3	0	0	0.1319
T_4	0	0	0
Cross Validation Deviance	1.208		

From the final LASSO model result, the following analysis conclusions can be made:

1. The first detour pattern (avoiding freeways) is largely dependent on OD distance (S_1). The negative sign of S_1 coefficient suggests that a longer OD distance is associated with a lower possibility of exhibiting the first detour pattern. On the other hand, whether a user can avoid freeway is highly correlated with trip distance. In other words, for shorter trips, travelers are usually more inclined to avoid freeway, while for longer trips freeway are less avoided. This observation is intuitive and matches with our common sense that longer travels rely more on freeway network.
2. The second detour pattern (using other freeways) appears more complicated with 4 features selected as the order: T_2, T_1, S_1, S_3. The coefficients T_2 and T_1 indicate that drivers who exhibit this pattern usually have longer travel times and shorter estimated travel times. In other words, the users who experienced longer travel time but are now notified of a new travel option with shorter estimated travel time are more likely to detour to other freeways. The positive coefficients S_1, S_3 also validate the previous suspicion that drivers exhibit this pattern tend to have longer travel distance.
3. The third detour pattern (O or D in downtown and frequently leave MoPac earlier or enter MoPac later) is most related with S_4 (OD's distance to construction area), i.e. if the user origin or destination location is close to the capacity reduction location, the possibility of them using a different ramp to enter or leave MoPac freeway and avoiding construction site will be high. The other significant features are T_3, S_1 and S_2. Another interesting finding is that drivers tend to detour with this pattern more on weekends (indicator T_3), which is likely due to the light traffic conditions on weekends, at which time using urban roads to bypass the construction area (but not completely leaving MoPac) is a good choice.

To summarize, the LASSO model built in this research is not only able to select proper variables and build a robust model, but is also capable of effectively explaining the underlying reasons for the three detour patterns identified by FDA clustering method. The previous suspicions stated in

section 5.2 are also validated by the result presented in Table 2. This indicates the proposed framework that firstly identifies detour patterns from smartphone trajectory data with a FDA approach, and then explains detour behavior choices with LASSO statistics model is promising in generating satisfactory results.

6 CONCLUDING REMARKS

To investigate the impacts of pre-planned road capacity reduction events and the heterogonous behavioral responses, this paper documents the usage of high-resolution vehicle trajectory data collected via a smartphone app, which is capable of keeping track of individual driver's behavior before and after road capacity reduction. A functional data analysis approach based clustering method is used to cluster trajectory data and identify detour patterns, and two logistic and a LASSO regression models are used to explain drivers' detour behavior choice for each pattern with spatial and temporal features of interest. A case study based on a lane closure event on MoPac Expressway in Austin, TX is used as an example, which demonstrates that: 1) the freeway capacity reduction triggered heterologous behavior responses, 2) driver detour behavior exhibits three major patterns, and 3) each detour pattern highly depends on spatial features such as trip length, distance to freeway entrance and distance to other alternative freeways, in addition to the temporal features when the trip happens. We hope this study can help researchers and practitioners gain a solid understanding of the traffic impact of such events as well as the behavior changes at the individual driver level, which is critical for agencies to foresee outcomes and take corresponding strategies so as to better manage traffic congestions.

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